

5 Experimental Evaluation

We evaluate our algorithm on several domains using both real and synthetic data distributions. We test the following algorithms: **Threshold Whittle** is the algorithm developed in this paper. **Qian et al. [22]**, a slow, but precise general method for computing the Whittle index, is our main baseline that we improve upon. **Random** selects k process to act on at random each round. **Myopic** acts on the k processes that maximize the expected reward at the immediate next time step. Formally, at time t , this policy picks the k processes with the largest values of $\Delta b_t = (b_{t+1}|a=1) - (b_{t+1}|a=0)$. **Oracle** fully observes all states and uses Qian et al. [22] to calculate Whittle indices. We measure performance in terms of *intervention benefit*, where 0% corresponds to the reward of a policy that is always passive and 100% corresponds to Oracle. All results are averaged over 50 independent trials.

5.1 Real Data: Monitoring Tuberculosis Medication Adherence

We first test on tuberculosis medication adherence monitoring data, which contains daily adherence information recorded for each real patient in the system, as obtained from Killian et al. [13]. The “good” and “bad” states of the arm (patient) correspond to “Adhering” and “Not Adhering” to medication, respectively. State transition probabilities are estimated from the data. Because this data is noisy and contains only the adherence records and not the intervention (action) information (as the authors state), we perturb the computed average transition matrix by reducing (increasing) $P_{\omega,1}$ by $\delta_1, \delta_2 (\delta_3, \delta_4)$ to obtain $P_{\omega,1}^p (P_{\omega,1}^a)$ for the simulation. Reward is measured as the undiscounted sum of patients (arms) in the adherent state over all rounds, where each trial lasts $T = 180$ days (matching the length of first-line TB treatment) with N patients and a budget of k calls per day.

In Fig. 4a, we plot the runtime in seconds vs the number of patients N . Fig. 4b compares the intervention benefit for $N = 100, 200, 300, 500$ patients and $k = 10\%$ of N . In the $N = 200$ case, the runtimes of a single trial of Qian et al. and Threshold Whittle index policy are 3708 seconds and 3 seconds, respectively, while attaining near-identical intervention benefit. Our algorithm is thus 3 orders of magnitude faster than the previous state of the art without sacrificing performance.

We next test Threshold Whittle as the resource level k is varied. Fig. 4c shows the performance in the $k = 5\%N$, $k = 10\%N$ and $k = 15\%N$ regimes ($N = 200$). Threshold Whittle outperforms Myopic and Random by a large margin in these low resource settings. We also affirm the robustness of our algorithm to δ , the perturbation parameter used to approximate real-world $P_{\omega,1}^p$ and $P_{\omega,1}^a$ from the data and present the extensive sensitivity analysis in Appendix G. Finally, in Appendix F we couple our algorithm to a Thompson Sampling-based learning approach and show it performs well in the real-world case where transition probabilities would need to be learned online, supporting the deployability of our work.

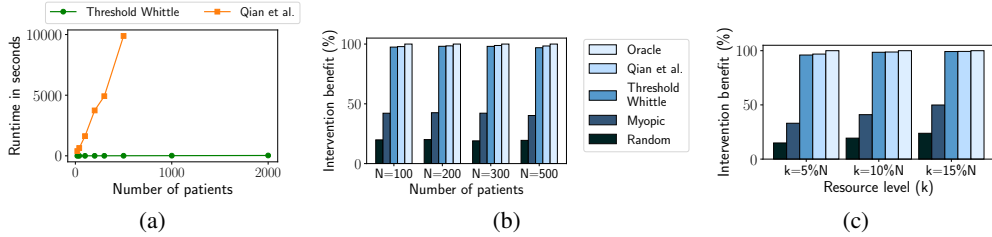


Figure 4: (a) Threshold Whittle is several orders of magnitude faster than Qian et al. and scales to thousands of patients without sacrificing performance on realistic data (b). (c) Intervention benefit of Threshold Whittle is far larger than naive baselines and nearly as large as Oracle.

5.2 Synthetic Domains

We test our algorithm on four synthetic domains, that potentially characterize other healthcare or relevant domains, and highlight different phenomena. Specifically, we: (i) identify situations when Myopic fails completely while Whittle remains close to optimal, (ii) analyze the effect of latent state entropy on policy performance, (iii) identify limitations of Threshold Whittle by constructing processes for which Threshold Whittle shows separation from Oracle, and (iv) test robustness of

our algorithm outside of the theoretically guaranteed conditions. To facilitate comparison with the real data distribution, we simulate trials for $T = 180$ rounds where reward is the undiscounted sum of arms in state 1 over all rounds. We consider the space of transition probabilities satisfying the assumed natural constraints, as outlined in Sec. 3.

Fig. 5a demonstrates a domain characterized by processes that are either self-correcting or non-recoverable. Self-correcting processes have a high probability of transitioning from state 0 to 1 regardless of the action taken, while non-recoverable processes have a low chance of doing so. We show that when the immediate reward is larger for the former than the latter, Myopic can perform even worse than Random. That is because a myopic policy always prefers to act on the self-correcting processes per their larger immediate reward, while Threshold Whittle, capable of long-term planning, looks to avoid spending resources on these processes. In this regime, the best long-term plan is to always act on the non-recoverable processes to keep them from failing. Analytical explanation of this phenomenon is presented in Appendix E. We set the resource level, $k = 10\%N$ in our simulation for Fig. 5a. Note that performance of Myopic drops as the fraction of self-correcting processes becomes larger and reaches a minimum at $x = 100\% - k = 90\%$. Beyond this point, Threshold Whittle can no longer completely avoid self-correcting processes and the gap subsequently starts to decrease.

Fig. 5b explores the effect of uncertainty in the latent state on long-term planning. For each point on the x -axis, we draw all transition probabilities according to $P_{\omega,1}^p, P_{\omega,1}^a \sim [x, x + 0.1]$. The entropy of the state of a process is maximum near 0.5 making long term planning most uncertain and as a result, this point shows the biggest gap with Oracle, which can observe all the states in each round. Note that Myopic and Whittle policies perform similarly, as expected for (nearly) stochastically identical arms.

Fig. 5c studies processes that have a large propensity to transition to state 0 when passive and a corresponding low active action impact, but a significantly larger active action impact in state 1. This makes it attractive to exclusively act on processes in the 1 state. This simulates healthcare domains where a fraction of patients degrade rapidly, but can recover, and indeed respond very well to interventions if already in a good state. To simulate these, we draw transition matrices with $P_{0,1}^p, P_{1,1}^p, P_{0,1}^a \sim [0.3, 0.32]$ and $P_{1,1}^a \sim [0.7, 0.72]$ in varying proportions and sample the rest from the real TB adherence data. Because the best plan is to act on processes in state 1, both Myopic and Whittle act on the processes with the largest belief giving Oracle a significant advantage as it has perfect knowledge of states.

Although we provide theoretical guarantees on our algorithm for forward threshold optimal processes with non-increasing belief, Fig. 5d reveals that Alg. 1 performs well empirically even with these conditions relaxed. Here, we sample processes uniformly at random from the state transition probability space, and use rejection sampling to vary the proportion of threshold optimal processes. Threshold Whittle performs well even when as few as 20% of the processes are forward threshold optimal; we briefly analyze this phenomenon in Appendix H.

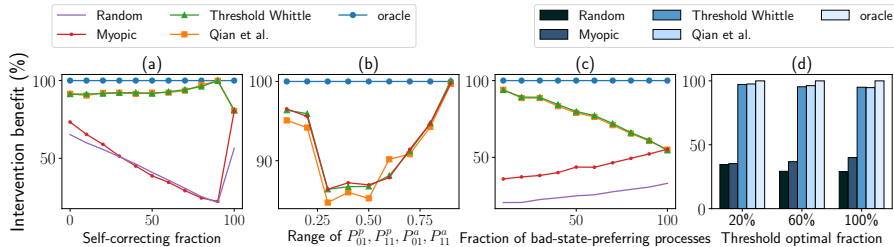


Figure 5: (a) Myopic can be trapped into performing even worse than Random while Threshold Whittle remains close to optimal. (b) Long-term planning is least effective when entropy of states is maximum. (c) Myopic and Whittle planning become similar when more processes are prone to failures. (d) Threshold Whittle is surprisingly robust to processes even outside of theoretically guaranteed conditions.

6 Conclusion

We open a new subspace of Restless Bandits, *Collapsing Bandits*, which applies to a broad range of real-world problems, especially in healthcare delivery. We give new theoretical results that cover a